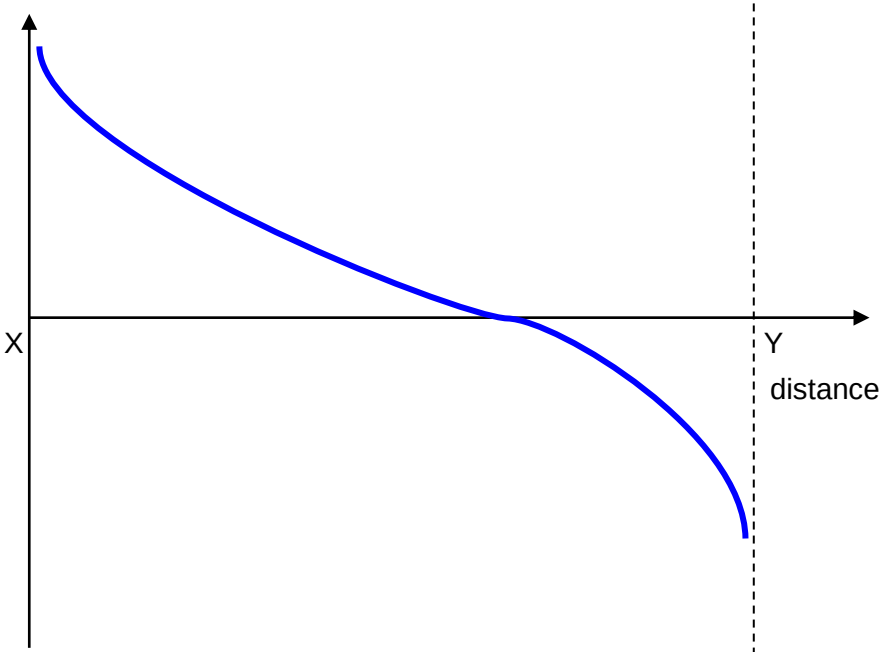


Qns	Answer	Marks
3(a)(i)	Since the direction of field strength is directed upwards at point O, the horizontal component of the electric field strength due to the 2 charges at O are equal in magnitude and opposite in direction. $\frac{q_1}{4\pi\epsilon_0 r^2} \sin 30^\circ = \frac{q_2}{4\pi\epsilon_0 r^2} \sin 60^\circ$ $\frac{q_1}{q_2} = \frac{\sin 60^\circ}{\sin 30^\circ} = 1.73$ $= 1.7$	1
3(a)(ii)	 correct shape, graph does not touch the y-axis or dotted line (else max 1) $E = 0$ at position nearer to Y Relative field strength at X larger in magnitude than at Y	1 1
3(b)(i)	electric field strength is a vector quantity Since the resultant field strength at point O is directed upwards, the resultant electric field strength is the <u>vector sum</u> of the <u>vertical component of the electric field strength</u> due to the 2 charges. $E_{\text{resultant}} = E_1 + E_2$ $= \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{r^2} \right) \cos 30^\circ + \frac{1}{4\pi\epsilon_0} \left(\frac{q_2}{r^2} \right) \cos 60^\circ$ $= \frac{q_2}{4\pi\epsilon_0 r^2} [(1.7) \cos 30^\circ + \cos 60^\circ]$ $= \frac{200 \times 10^{-9}}{4\pi (8.85 \times 10^{-12}) (0.50)^2} [(1.7) \cos 30^\circ + \cos 60^\circ]$ $= 1.42 \times 10^4 \text{ V m}^{-1}$	1 1

Qns	Answer	Marks
(b)(ii)	electric potential is a scalar quantity total electric potential at point O is <u>sum</u> of <u>electric potential at that point due to the 2 charges</u>. $V_{\text{resultant}} = V_1 + V_2$ $= \frac{1}{4\pi\epsilon_0} \left(\frac{q_1}{r} \right) + \frac{1}{4\pi\epsilon_0} \left(\frac{q_2}{r} \right)$ $= \frac{q_2}{4\pi\epsilon_0 r} [(1.7) + 1] = \frac{200 \times 10^{-9}}{4\pi(8.85 \times 10^{-12})(0.50)} (2.7)$ $= 9710 \text{ V}$	1
(b)(iii)	work done $W = q\Delta V$ $= (-2q_2)(V_{\text{final}} - V_{\text{initial}})$ $= (-2)(200 \times 10^{-9})(9710 - 0)$ $= -0.00388 \text{ J}$ $= -2.43 \times 10^{16} \text{ eV}$	1 1

Qns	Answer	Marks
4(e)	Force per unit current unit length of wire	